

Yokohama City University - Quantitative Text Analysis in Social Sciences

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シラバス参照

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シラバス参照

科目名

Course name

Topics in Economics B

担当教員

Instructor

御器谷 裕樹

単位数

No of credits

2

期間・曜日・時限

Term/day/period

前期後半 水曜日 4時限 YCUスクエア403教室

前期後半 水曜日 5時限 YCUスクエア403教室

コース

・専攻名

Faculty

Quantitative Text Analysis in Social Sciences

学修到達

目標

Course

objective

By the end of this course, students will have developed the ability to formulate research questions relevant to the social sciences and conduct rigorous research using quantitative text analysis methods. Students will gain a theoretical understanding of the methodology, learn to select suitable techniques for various research questions, and write R code to implement text analysis. Furthermore, students will develop the skills necessary to interpret results accurately and effectively within their specific research contexts.

授業概要・形態

Lecture Outline

and Class Format

In recent years, quantitative text analysis has become increasingly utilized across various disciplines within the social sciences, including economics, political science, and sociology. This method, which involves extracting insights by quantitatively analyzing textual data, has roots in fields such as computational linguistics and machine learning. Given the diversity of approaches available, a deep understanding of the methodology and its appropriate application is essential.

This course will focus on applying quantitative text analysis within the social sciences. Students will learn about the historical development, theoretical foundations, and computational techniques behind these methods. The primary programming language used in this course will be R, with a particular emphasis on the `quanteda` package for conducting text analysis.

成績評価

方法

Grading

methods

40% Homework assignments

40% Final paper (Analyzing a social science topic using quantitative text analysis)

20% In-class presentation

The presentation and the paper will cover the same topic. The presentation will focus on descriptive results, while the paper will emphasize in-depth analysis and theoretical implications.

This is an introductory-level course. Students who demonstrate a solid understanding of the material and are able to independently apply the core methods (approximately 70% of the course content) will receive the highest evaluation.

前提科目

Prerequisite

courses

No prior knowledge of R or quantitative text analysis is required. This course is designed for beginners. Thorough support will be provided. We strongly encourage students who are new to programming or those who may have concerns about their English proficiency to enroll without hesitation.

事前事後学修

及び学修上・

履修申請上の

留意点

Pre- and

post-study

and points

to keep in mind

when studying

and applying

for courses

Students are required to bring their laptops to class and will engage in hands-on exercises, applying the methods covered in the lectures.

教科書・

参考図書等

Textbooks,

reference books,

etc.

1. Grimmer, J., Roberts, M. E., & Stewart, B. M. (2022). Text as data: A new framework for machine learning and the social sciences. Princeton University Press.
2. Benoit, K., Watanabe, K., Wang, H., Nulty, P., Obeng, A., Müller, S., & Matsuo, A. (2018). quanteda: An R package for the quantitative analysis of textual data. Journal of Open Source Software, 3(30), 774-774.
3. Wickham, H., Çetinkaya-Rundel, M., & Golemund, G. (2023). R for data science. " O'Reilly Media, Inc.".

教員研究室

Faculty

Laboratory

教員オフィス

アワー

Office

Hours

ホーム

ページ

Homepage

教員の

実務経験

について

Work

Experience

of Faculty

アクティブ

ラーニング

について

Active

Learning

Each class will combine lectures with computer programming exercises. Students are also expected to engage in discussions with their peers regarding their analyses.

類型1) 講義中心アクティブラーニング

持続可能な

開発目標 (SDGs)

17の目標との関連性

Relevance

to the 17

Sustainable

Development

Goals(SDGs)

This course contributes to SDG 4: Quality Education by promoting advanced, equitable, and inclusive learning opportunities through the integration of computational methods into social science research. By equipping

students with skills in quantitative text analysis, the course enhances their analytical capabilities and critical thinking in an interdisciplinary context. The use of programming (R) and data-driven methodologies allows students to engage with complex problems, preparing them for research and innovation that address pressing social issues. This aligns with the aim of SDG 4 to ensure inclusive and equitable quality education and to promote lifelong learning opportunities for all.

The course also aligns with SDG 9: Industry, Innovation, and Infrastructure, particularly in its focus on developing computational techniques that are foundational for innovation. Quantitative text analysis is a cutting-edge approach that leverages advances in machine learning and computational linguistics, fields that are essential for fostering innovation in both academic research and industry. By training students in these methods, the course contributes to the development of sustainable infrastructures for data analysis and decision-making, supporting the broader goal of building resilient infrastructure, promoting inclusive and sustainable industrialization, and fostering innovation as outlined in SDG 9.

授業計画

(週単位)

Course

plans

Class 1: Introduction: Quantitative Text Analysis in Social Sciences

- Lecture: Overview of QTA. How "Text as Data" transforms social science research.
- Lab: Setting up R and RStudio. Basic R syntax and project management.

Class 2: R Programming and Data Handling

- Lecture: Data structures (vectors, data frames) and the tidyverse workflow for social data.
- Lab: Importing and cleaning text datasets (CSV, JSON, and web-scraped data).

Class 3: Foundations with Quanteda: Corpora and Tokens

- Lecture: Managing large-scale text collections. The logic of tokenization in social science.
- Lab: Creating and subsetting corpora using the quanteda package.

Class 4: Feature Extraction: Document-Feature Matrix (DFM)

- Lecture: Preprocessing (stopwords, stemming) and the vector space model.
- Lab: Constructing and weighting DFMs (tf-idf). Understanding sparsity in social discourse.

Class 5: Descriptive Analysis and Dictionary Methods

- Lecture: Visualizing text data and the logic of dictionary-based sentiment analysis.
- Lab: Word clouds, frequency plots, and implementing custom dictionaries for social indicators.

Class 6: Quantitative Scaling I: Wordscores and Wordfish

- Lecture: Estimating ideological positions. Supervised (Wordscores) vs. Unsupervised (Wordfish) scaling.
- Lab: Mapping political or economic positions using classic social science datasets.

Class 7: Quantitative Scaling II: Correspondence Analysis

- Lecture: Dimensionality reduction and Correspondence Analysis in text analysis.
- Lab: Visualizing relationships between authors, documents, and words in a low-dimensional space.

Class 8: Automated Classification I: Naive Bayes and Newsmap

- Lecture: The Naive Bayes classifier and semi-supervised extensions like Newsmap for geographical/thematic classification.
- Lab: Training a classifier and using Newsmap to categorize news or policy documents.

Class 9: Automated Classification II: Regularized Regression

- Lecture: Regularized regression classifiers (Lasso/Ridge/Elastic Net). Feature selection in high-dimensional data.
- Lab: Predicting document labels and interpreting the most influential "social" keywords.

Class 10: Topic Modeling I: Latent Dirichlet Allocation (LDA)

- Lecture: Introduction to Topic models. Discovering latent themes in large-scale social text.
- Lab: Running LDA, selecting the number of topics, and interpreting results.

Class 11: Topic Modeling II: Structural Topic Model (STM)

- Lecture: Advanced topic modeling. Incorporating social metadata (time, gender, party) into STM.
- Lab: Analyzing how topics shift across different demographic or temporal groups.

Class 12: Word Embeddings I: Word2vec

- Lecture: The "Distributional Hypothesis." Introduction to word embeddings and Word2vec.
- Lab: Training embeddings and identifying semantic similarities in social science contexts.

Class 13: Word Embeddings II: Latent Semantic Scaling (LSS)

- Lecture: Combining embeddings with scaling. The theory and application of Latent Semantic Scaling (LSS).
- Lab: Using LSS to measure specific concepts (e.g., "democracy" or "uncertainty") in text.

Class 14: Student Presentations: Research Projects

- Session: Students present their analysis of a social science topic using the methods learned.
- Discussion: Peer feedback on methodological choices and interpretation of results.

Class 15: Multilingual Analysis, Ethics, and Final Review

- Lecture: * Text analysis for Asian languages: Challenges in analyzing Japanese, Chinese, and Korean (CJK) data.
 - o Tokenization and morphological analysis for non-segmented languages.
 - o Ethical considerations: bias, privacy, and the impact of Large Language Models (LLMs) in social science.
- Lab: * Practical exercises: Handling CJK text data in R.
- o Course wrap-up and final consultation for the short paper.

備考

Remarks